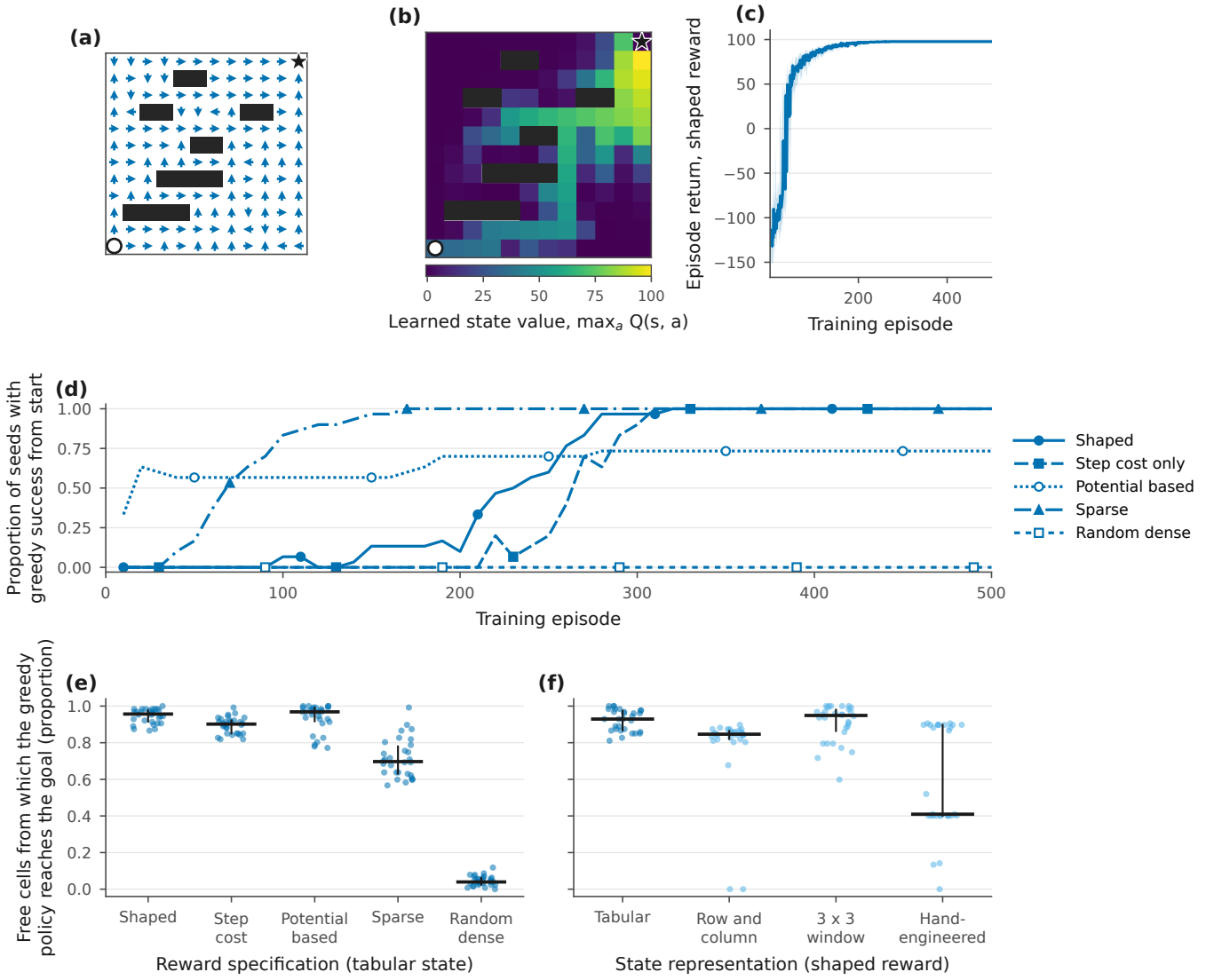

(a)
(b)
0
25
50
75
100
Learned state value, max$_a$ Q(s, a)
(c)
Episode return, shaped reward
100
50
0
−50
−100
−150
200
400
Training episode
(d)
Proportion of seeds with greedy success from start
1.00
0.75
0.50
0.25
0.00
0
100
200
300
400
500
Training episode
Shaped
Step cost only
Potential based
Sparse
Random dense
(e)
Free cells from which the greedy policy reaches the goal (proportion)
1.0
0.8
0.6
0.4
0.2
0.0
Shaped
Step cost
Potential based
Sparse
Random dense
Reward specification (tabular state)
(f)
Tabular
Row and column
3 x 3 window
Hand-engineered
State representation (shaped reward)